

Aditya Mukhopadhyay

AI Engineer – Badho Technologies
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EDUCATION

B.Tech, Information Technology

IGEC Sagar (RGPV), Madhya Pradesh
2021 – 2025 · CGPA 7.5

Class XII (PCM)

SJC S.S. School, Ranjhi, Jabalpur
2021 · 93.4%

LINKS

Portfolio

adityamukhopadhyay.vercel.app

LinkedIn

adityamukhopadhyay2003

GitHub

adityamukhopadhyay

SKILLS

Core

Python · TypeScript · SQL · GraphQL
(Hasura) · FastAPI · PostgreSQL ·
Supabase · Docker · AWS (EC2,
Lambda, S3) · Railway · Vercel

AI / LLM

Claude Agent SDK · Model Context
Protocol (MCP) · LangGraph · Claude
/ GPT / Gemini APIs · FAISS ·
embeddings & RAG · prompt
caching & cost optimization · agent
guardrails & evals · n8n

PROJECTS

Aditya MCP – agent-ops infrastructure

2026

Work across many AI coding sessions
was hard to report. Read-only **MCP
server** over session transcripts, git
history and handoff docs; powers a
daily work report and stale-doc
nudges.

Serverless Multimodal Call QA

2025

Call QA sampled a few calls by hand.
AWS Lambda feeds audio straight to
Gemini 2.5 Pro with campaign
questions, writing strict JSON to
PostgreSQL/Metabase. **100% call
coverage**.

SUMMARY

AI Engineer who takes LLM systems from idea to production and keeps them running: multi-agent orchestration on the Claude Agent SDK, MCP servers with real auth, WhatsApp agents for retailers, and semantic search and matching pipelines – Python, TypeScript, PostgreSQL, n8n and AWS, with deterministic guardrails and cost design built in.

WORK EXPERIENCE

AI Engineer | Badho Technologies Pvt. Ltd.

Jun 2025 – Present

• Autonomous Delivery Fleet – a multi-agent tree per rider (Claude Agent SDK, MCP)

- Delivery problems surfaced only when someone noticed a rider was stuck. Built a fleet where a root **orchestrator agent** runs each rider's day (~20 concurrent) and delegates to **sub-agents** – an **empath** drafting every Hinglish message, a **driver-profile analyst** setting tone from history, an **escalation governor**, a **judgement tiebreaker**, a **Google Maps route planner** – over one in-process **MCP server**: trip snapshot, log deltas, telemetry, SQL, **WhatsApp**, **push**, **Freshdesk tickets**, **vision on rider photos**. Runs unattended for **13-hour shifts**, plus an **"Ask the fleet"** agent for operators; guards below the model, **23 offline test suites**.

• Buyer MCP – the buyer app as 100 agent tools (TypeScript, Hasura GraphQL)

- Agents had no safe way to act inside the buyer app. Wrote an MCP server exposing it as **100 typed tools** – **product search**, **carts**, **coupons** and **schemes**, **cart deeplinks**, **orders** and **cancellations**, **wallet** and **vouchers**, **support tickets** – behind phone-OTP OAuth and **per-buyer JWTs**, importing the app's own cart and coupon functions so prices never differ. Serves the production sales agent, Claude Desktop and Claude Code.

• WhatsApp AI Sales Agent (Claude Agent SDK + Buyer MCP)

- Retailers ask questions and abandon carts on WhatsApp faster than a sales team can answer. Shipped a live agent that **searches the catalogue**, **builds carts**, **applies coupons**, **sends order deeplinks**, **answers order** and **cancellation questions from live state**, **handles complaints and images**, and **raises support tickets**: a tool-using worker reasons, a tool-less reply agent is the only voice, deterministic gates stop it claiming an unplaced order or filing duplicate tickets. Handles buyer chats end to end in production.

• Courier Ops Autopilot – claims & tickets (FastAPI, remote MCP, Claude Haiku)

- Weight overcharges and failed deliveries with the courier partner went unclaimed; filing was manual. Built a backend, dashboard and remote MCP over the courier's API that **auto-files weight claims with vision over sorter images**, **raises NDR/lost/damaged tickets** and **drafts vendor replies**; one liveness gate halts every loop and model call if the vendor session dies. **698 claims and 370 tickets filed in 24 hours; 2,143 closed** to date.

• Search, Matching & Catalog Intelligence (FAISS, HDBSCAN, n8n, PostgreSQL)

- Retailers search in Hinglish and phonetic spellings, **85,000+ SKUs** had no variant grouping, and **170,000+ competitor SKUs** needed mapping to the catalog. Built sub-100 ms hybrid search (FAISS + local embeddings + PostgreSQL metaphone), sister-product clustering (MiniLM embeddings, HDBSCAN/K-Means) and an n8n matching funnel ending in a GPT-4o-mini decision. **~40,000 product groups** mapped, **170,000+ SKUs** matched.

• Order-Lifecycle WhatsApp Notifications (event-driven, PostgreSQL, Supabase)

- Buyers heard nothing as their orders moved. Built an event-driven notification engine over the read replica – pollers per stage, an idempotent send ledger, event-time guards, per-flow scope gates and boot-time template checks – sending WhatsApp updates through approved templates. **10 order stages, ~500 messages a day** to the whole buyer base.

• Live Money Paths – referral credits & QR cashback (Next.js, Supabase, PostgreSQL)

- Reward programmes on real money must never pay twice or pay early. Built a referral engine (buyer webview with app-injected auth, rewards released only on the referred buyer's first delivered order) and carton-QR cashback where double credit is blocked by a **partial unique index on the wallet ledger**. **407 buyers referred, INR 48.1K paid out, 114 orders** from referred buyers and **155 orders (INR 1.03L)** across both cohorts since referral.

AI Developer (Freelance) | Strix Digital

Jul 2023 – May 2025

- Small clients wanted AI features without an in-house team. Built AI web apps (a cold-email generator on **Groq**), scrapers and automation tooling, and shipped an iOS app for a Houston client.

Data Science Intern | Acmegrade (in collab. with IIT Bombay)

Apr 2022 – Jun 2022

- Built ML models for **movie recommendation**, **sales forecasting** and **CNN sound classification**; EDA and feature engineering on multi-modal datasets.